

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

Honour School of Engineering, Economics and Management (Part II)

Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2015

Friday 30th May 2014, 09:30 - 12:30

Please start the answer to each question on a separate page.

There are 11 questions in this paper.

*Answer **three** short questions from Part A and **two** long questions from Part B.*

Part A attracts 1/3 of the marks in total; each question in part B attracts 1/3 of the marks.

For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

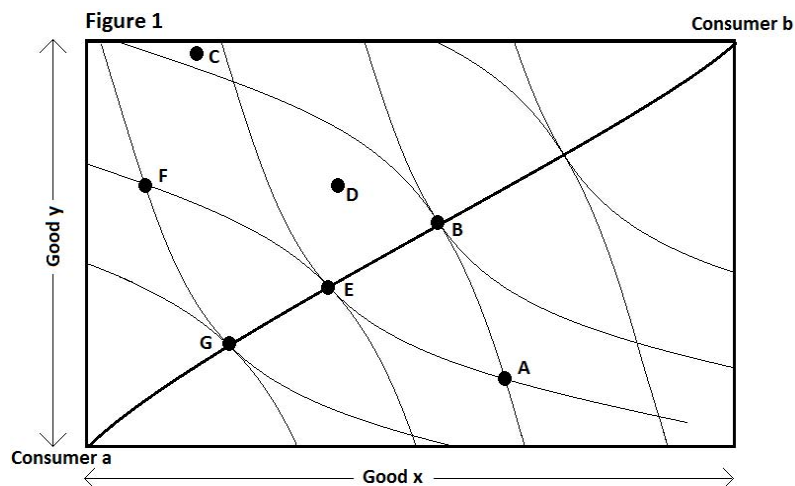
Part A

1. (a) Explain the Pareto Criterion for choosing between social outcomes. What are its advantages and disadvantages? [15%]

Definition from lectures: A social outcome A is Pareto superior to B if no-one finds A worse than B , and at least one person finds it strictly better.

Advantages? Can argue that the ordering it provides is not controversial: if $A \succ B$ in the Pareto sense, it would hard to claim that A should not be socially preferred, except when individuals really do not know what is good for them (e.g. child doesn't want to go to school) in which case we might adopt a paternalistic criterion. Candidates might try to argue that giving more to the richest person wouldn't be a social improvement, but if so they need to be clear about what the negative effect is (if there were negative externalities it would not be a Pareto improvement).

Disadvantages? Partial ordering, probably best illustrated by an example. A and B may be non-comparable, whether or not they are Pareto efficient.



The Edgeworth Box in Figure 1 represents all possible allocations between consumers a and b of two goods x and y in an exchange economy. The consumers have convex preferences. Indifference curves and the contract curve are shown.

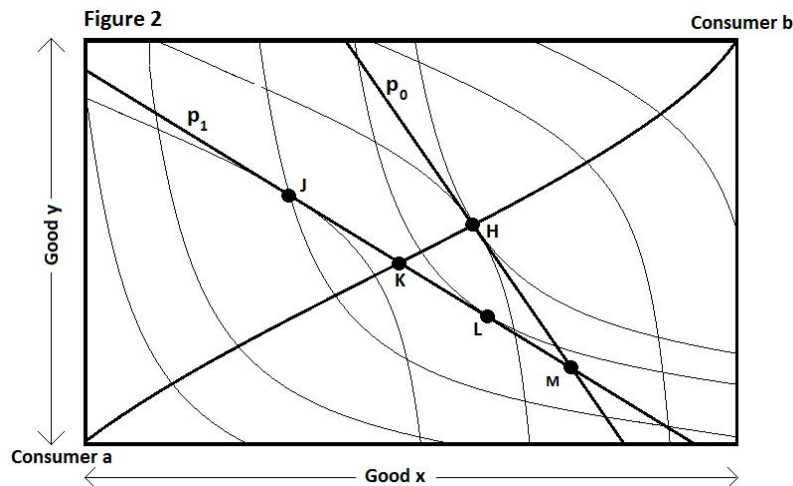
- (b) Which of the labelled points in Figure 1 are Pareto efficient? Which are Pareto superior to point A ? Which are Pareto inferior to A ? [20%]

Efficient: B, E, G . Superior to A : B, E, D . Inferior to A : C

- (c) “Point G is Pareto efficient and Point A is not, but G is not Pareto superior to A.” Is this statement true? Explain your answer. Suggest a criterion that a social planner could use to choose between A and G. [20%]

Yes, it is true. Define Pareto efficiency. Then, in the diagram, a and b both have convex preferences; G is at a tangency point so the set of allocations that person a prefers to G does not intersect with the set that b prefers to G. So G is efficient; A is not because in this case the two sets do intersect. However, person a strictly prefers A to G.

To choose between A and G, social planner needs some kind of social welfare function. A good answer might raise the issue of comparability of utility. e.g. if SP can compare utility levels between a and b, a Rawlsian criterion could be used. If utility is cardinal and fully comparable, a utilitarian SWF could be used.



In Figure 2, M is the initial endowment. Let the price of good y be 1. The straight lines correspond to two possible prices p_0 and p_1 for good x .

- (d) Explain carefully why p_0 represents a competitive equilibrium price ratio, and p_1 does not. Use the terms *excess demand* and *excess supply* in your answer. [25%]

Competitive equilibrium $\equiv$ price-taking + optimisation + market clearing. When the price ratio is p_0 , both consumers choose the bundle at H, where their MRS between the goods is equal to the price ratio. So the markets for both goods clear. This is not the case with p_1 : person a would choose point L, and person b would choose J; there would be excess demand for good x and excess supply of good y .

- (e) Suppose that the social planner prefers point K to point H, and is able to redistribute good x but not good y . Explain what might be done to achieve allocation K. Would the equilibrium price ratio be greater or less than p_1 ? [20%]

K is Pareto efficient (it is on the contract curve), and preferences are convex, so there is an equilibrium price ratio $p_e > p_0$, equal to the MRSs of the two consumers at K. The social planner would redistribute some of good x from person a's endowment to person b's endowment, so that at price p_e both consumers could afford to buy bundle K. Diagrammatically the new endowment point would be on a steeper line through K, with the same y -coordinate as M.

2. Suppose that two firms produce horizontally differentiated products. The firms compete for a single period and choose prices simultaneously. Each firm has the same constant marginal cost of production $c \geq 0$, and fixed costs are zero. The linear demand functions are given by:

$$\begin{aligned} q_1(p_1, p_2) &= 100 - \alpha p_1 + \beta p_2 \\ \text{and } q_2(p_1, p_2) &= 100 - \alpha p_2 + \beta p_1. \end{aligned}$$

Assume that $\alpha > \beta > 0$ and that $100 > \alpha c$.

- (a) Explain the difference between horizontally differentiated and vertically differentiated products, giving examples of each. Explain why the demand functions above represent horizontally differentiated products. [25%]

Vertical differentiation: Goods vary in quality. If the goods sell at the same price, everybody chooses the higher quality variety. e.g. Medical care.

Horizontal differentiation: Goods vary on dimensions other than quality. If the goods sell at the same price, consumers differ in their ranking. e.g. Breakfast cereals

When $p_1 = p_2$ both firms have the same amount of demand, but firm i retains some demand when $p_i > p_j$, and so the products are horizontally differentiated.

- (b) Show that the symmetric pure-strategy Bertrand-Nash equilibrium prices are:

$$p_i^* = p_j^* = \frac{100 + \alpha c}{2\alpha - \beta}$$

Find the corresponding quantities, and check that the prices and quantities are positive at the equilibrium. [25%]

$$\begin{aligned} \pi_i &= (100 - \alpha p_i + \beta p_j)(p_i - c) \\ \text{FOC: } \frac{\partial \pi_i}{\partial p_i} &= 100 - 2\alpha p_i + \beta p_j + \alpha c = 0 \\ \Rightarrow p_i^*(p_j) &= \frac{100 + \beta p_j + \alpha c}{2\alpha} \\ \text{SOC: } \frac{\partial^2 \pi_i}{\partial p_i^2} &= -2\alpha < 0 \end{aligned}$$

Solve the BR functions simultaneously. Since we are searching for a symmetric equilibrium

$$\begin{aligned} p_i^*(p_j) &= \frac{100 + \beta p_i + \alpha c}{2\alpha} \\ \Rightarrow 2\alpha p_i^* &= 100 + \beta p_i + \alpha c \\ \Rightarrow p_i^* = p_j^* &= \frac{100 + \alpha c}{2\alpha - \beta} \end{aligned}$$

Clearly, $p_i^* > 0$ since $\alpha > 0$, $c \geq 0$, so $100 + \alpha c > 0$, and $\alpha > \beta > 0$, so $2\alpha - \beta > 0$.

$$\begin{aligned}
 q_i^* &= 100 - \alpha p_i^* + \beta p_j^* \\
 &= 100 - (\alpha - \beta) \left(\frac{100 + \alpha c}{2\alpha - \beta} \right) \\
 &= \frac{100\alpha - (\alpha - \beta)(\alpha c)}{2\alpha - \beta} \\
 &= \frac{\alpha(100 - \alpha c + \beta c)}{2\alpha - \beta} > 0
 \end{aligned}$$

given that $100 > \alpha c$, $\alpha > \beta > 0$ and $c \geq 0$.

- (c) Show how the equilibrium prices change in the parameters α, β and c . Provide a brief economic explanation of these comparative statics results. [25%]

$$\begin{aligned}
 \frac{\partial p_i^*}{\partial \alpha} &= \frac{c(2\alpha - \beta) - (100 + \alpha c)2}{(2\alpha - \beta)^2} = \frac{-c\beta - 200}{(2\alpha - \beta)^2} < 0 \\
 \frac{\partial p_i^*}{\partial \beta} &= \frac{(-1)(100 + \alpha c)(-1)}{(2\alpha - \beta)^2} = \frac{100 + \alpha c}{(2\alpha - \beta)^2} > 0 \\
 \frac{\partial p_i^*}{\partial c} &= \frac{\alpha}{2\alpha - \beta} > 0
 \end{aligned}$$

As α goes up, demand becomes more sensitive to the firm's own price, so the equilibrium price falls.

As β goes up, demand becomes more sensitive to the rival's price, so the equilibrium price rises.

As c goes up, the per unit production cost goes up, so the equilibrium price rises

- (d) Suppose now that the two firms merge. The new firm acts as a monopolist, choosing p_1 and p_2 . Find the profit-maximizing price assuming that the monopolist sets the same price for both products, that is $p_1 = p_2$. Is this price higher or lower than the equilibrium price from part (b)? [25%]

$$\begin{aligned}
 \pi &= 2(100 - \alpha p + \beta p)(p - c) \\
 \text{FOC} &: \frac{\partial \pi}{\partial p} = 2(100 - 2\alpha p + 2\beta p + \alpha c - \beta c) = 0 \\
 &\Rightarrow p^* = \frac{100 + \alpha c - \beta c}{2(\alpha - \beta)} \\
 \text{SOC} &: \frac{\partial^2 \pi}{\partial p^2} = -4\alpha + 4\beta < 0
 \end{aligned}$$

Finally, for an algebraic comparison of prices:

$$\begin{aligned}
 \frac{100 + \alpha c - \beta c}{2(\alpha - \beta)} &> \frac{100 + \alpha c}{2\alpha - \beta} \Leftrightarrow \\
 (100 + \alpha c)(2\alpha - \beta) - \beta c(2\alpha - \beta) &> (100 + \alpha c)(2\alpha - 2\beta) \Leftrightarrow \\
 (100 + \alpha c)\beta &> \beta c(2\alpha - \beta) \Leftrightarrow \\
 100 &> \alpha c - \beta c
 \end{aligned}$$

which holds given $100 > \alpha c$ and $\beta > 0$ and $c \geq 0$.

Alternatively a good economic explanation would be acceptable. e.g. The monopolist takes into account that raising the price of one product increases demand for the other, which competitors would not do - for them this is an externality.

3. An agent can exert two effort levels, $e = 1$ or 0 . When $e = 1$ a production error occurs with probability 0.25; when $e = 0$ the probability of a production error is 0.75. His utility is:

$$u(w, e) = 4 - \frac{5}{w} - 2e,$$

where w is the wage received. Production errors are observable and so can be introduced into the agent's contract, but effort cannot. The output produced is worth 12 if there is no error and 0 otherwise. The principal is risk neutral. Assume the agent's reservation utility is $\underline{u} = 0$.

- (a) Suppose the principal wishes the agent to undertake effort $e = 0$. What is the participation constraint in this case? What can you say about the incentive constraint? Find the wage schedule she will offer the agent, conditional on the observed error. What is the principal's expected profit? [35%]

The 2-effort, 2-outcome agency model was explained graphically in the lectures.

If the principal demands $e = 0$, there is no incentive constraint. The participation constraint is :

$$4 - (5/w) - 2e = 0 \quad \Rightarrow w = 1.25.$$

The principal's expected profit is $0.25 \times 12 - 1.25 = 1.75$

- (b) Suppose the principal wishes the agent to undertake effort $e = 1$. Write down the participation and incentive constraints. Find the wage schedule she will offer the agent, conditional on the observed error. What is the principal's expected profit in this case? [45%]

If the principal demands $e = 1$, she agrees to pay a wage w^S if the production outcome is a success (i.e., error-free) and w^F if the production is a failure. The participation and incentive constraints imply the following two equations:

$$\frac{3}{4w^S} + \frac{1}{4w^F} = \frac{2}{5}, \quad \text{and} \quad \frac{1}{w^F} - \frac{1}{w^S} = \frac{4}{5}.$$

The solution is $w^S = 5$ and $w^F = 1$. The expected profit of the principal is

$$\frac{3}{4}(12 - 5) + \frac{1}{4}(0 - 1) = 5$$

which is greater than that obtained with $e = 0$.

- (c) Find the agency cost involved in implementing effort $e = 1$. [20%]

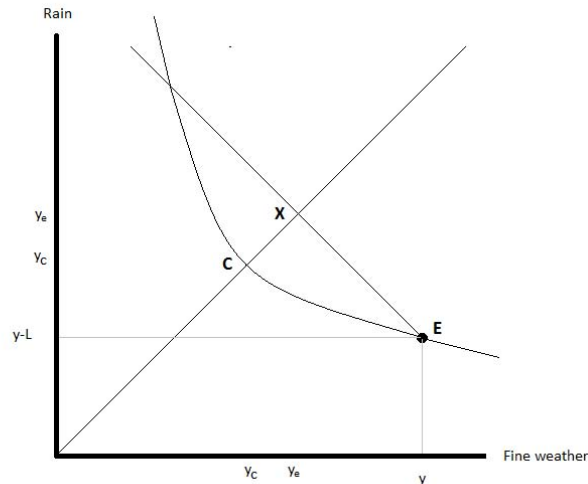
The certainty equivalent $\hat{w}$ corresponding to (w^S, w^F) solves:

$$\frac{1}{\hat{w}} = \frac{3}{4w^S} + \frac{1}{4w^F} = \frac{8}{20} \Rightarrow \hat{w} = 2.5$$

Hence, the agency cost is:

$$\begin{aligned} \text{Expected wage} - \hat{w} &= 3/4 \times 5 + 1/4 \times 1 - 2.5 \\ &= 4 - 2.5 = 1.5 \end{aligned}$$

4. Arthur is a risk-averse farmer whose income will be y if the weather is fine during harvest, and $y - L$ if it rains. The probability of rain is 0.5.
- (a) Draw a carefully-labelled 2-state diagram to illustrate: his endowment point; his expected income; his certainty-equivalent income. [25%]

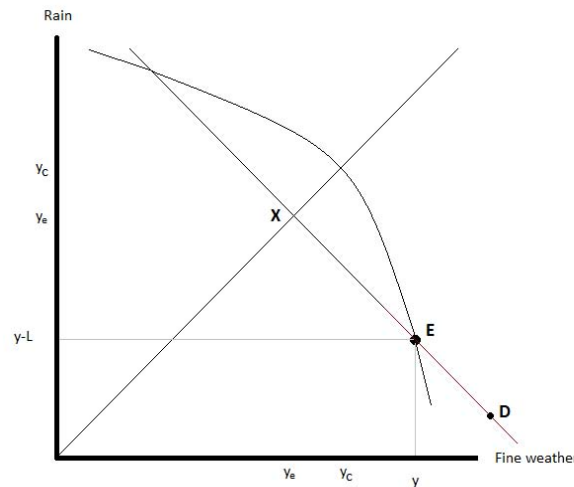


E is the endowment point; the line EX represents points of the same expected value (slope = -1 since probability = 0.5), y_e ; at X income is equal to y_e in both states. Indifference curves are convex (risk-aversion); point C is on the same indifference curve as E but income is y_c in both states - y_c is certainty equivalent income.

- (b) Use the diagram to illustrate and explain (i) how he would benefit from buying insurance in a competitive insurance market; (ii) the maximum profit that his risk-neutral neighbour Bertha could make by offering him full insurance. [25%]

In a competitive market insurers would offer fair insurance (making zero profit); this would allow Arthur to achieve any point on the line EX. He reaches the highest possible indifference curve at point X, which gives full insurance: income $y_e = y - 0.5L$ in both states. If Bertha could make a take-it-or-leave-it offer, she could provide insurance guaranteeing Arthur income y_c in each state. Her profit would be the risk premium $y_e - y_c$.

- (c) Suppose instead that Arthur was risk-seeking. Describe, and illustrate on a 2-state diagram, an agreement with Bertha that would make Arthur better off, and that Bertha would be just willing to accept. [20%]



Now Arthur has convex indifference curves. Bertha is indifferent between E and all other points on EX. Arthur can get to a higher indifference curve by moving along EX to a more risky point, such as D. Here, Bertha pays Arthur amount z in fine weather, and Arthur pays Bertha z if it rains.

Now suppose Arthur and Bertha are both risk-averse farmers with utility of income, y , given by $u(y) = \ln y$. On each farm, income will be 8 if it is fine, and 2 if it rains.

As before, the probability of rain on each farm is 0.5. Furthermore, with probability $p \leq 0.5$, both will have fine weather, with probability p it will rain on both farms, and otherwise just one of them will have rain.

- (d) Find the increase in utility that each of them could achieve by pooling their incomes, as a function of p . How do the benefits of pooling change as p increases? Explain intuitively why this happens. [30%]

Without pooling, each farmer has $EU = 0.5 \ln 8 + 0.5 \ln 2 = \ln 4$ ($= 1.386$).

With pooling: $EU = p \ln 8 + p \ln 2 + (1 - 2p) \ln 5 = 2p \ln 4 + (1 - 2p) \ln 5$.

Increase = $\Delta EU = (1 - 2p)(\ln 5 - \ln 4) = \ln 1.25(1 - 2p) = 0.223(1 - 2p)$

The change in EU decreases with p . The higher is p , the more highly correlated are the incomes on the two farms, and hence the lower the benefits of pooling. A very good answer might go further: the best case is $p = 0$, when they are perfectly negatively correlated and $\Delta EU = 0.223$; when $p = 0.5$ they are perfectly correlated and there is no gain from pooling, $\Delta EU = 0$. $p = 0.25$ represents the case of independence. A positive correlation ($p > 0.5$) seems likely if they are neighbours.

5. (a) What is the *tragedy of the commons*? Use the concepts of rivalry and excludability in your answer. [20%]

Common property resources, like fish stocks in international waters or common grazing land, are *rival*, or at least partially so, and non-excludable. No-one can be prevented from using them; but one person's use diminishes the value of the resource for others. Typically they are only partially rival, in that low-level use does not affect others negatively, but at higher levels of use there is a problem of congestion. Non-excludability may arise from a technological problem (the difficulty of exclusion or monitoring), or a legal restriction (the land is commonly-owned), or both.

A particular road is used by N drivers each day. Each driver obtains a benefit v from the journey, and incurs a cost $c(N)$ that increases with the number of drivers on the road, because of congestion. The cost incurred by each driver is given by:

$$c(N) = a + bN$$

- (b) Show that the equilibrium number of drivers using the road will be:

$$N_e = \frac{v - a}{b} \quad [10\%]$$

The net benefit of the road to each driver, when N drivers use the road, is: $v - c(N) = v - a - bN$. Each driver will choose to use it if and only if: $v - a - bN \geq 0$. In equilibrium $v = a + bN$; solving for N gives N_e .

- (c) Write down an expression for the social surplus, W (total benefits minus costs), provided by the road when it is used by N drivers. Find the number of drivers, N^* , at which the social surplus is maximised. What is the social surplus when $N = N_e$? Comment on your answers. [25%]

The total net benefit is $N \times \text{benefit-per-driver}$: $W = N(v - a - bN)$.

$$\frac{dW}{dN} = v - a - 2bN = 0 \quad \Rightarrow \quad N^* = \frac{v - a}{2b}$$

is the optimum number of drivers (half of the equilibrium number).

When $N = N_e$ $W = 0$ - drivers 'enter' until the individual net value of using the road is zero, so the social surplus is zero.

When $N = N^*$ $W = b(N^*)^2$ - it is optimal to restrict N so that the benefit to the marginal driver is equal to the marginal social cost.

- (d) Suppose the government charges a toll, t , (a congestion charge) for each journey. Find the optimum toll. Interpret your answer in terms of the externality that each driver imposes on others. [25%]

The toll should be equal to MSC-MPC at the optimum:

$$t = (a + 2bN^*) - (a + bN^*) = bN^* = (v - a)/2$$

Alternatively, the equilibrium with the toll is where $N_e = (v - a - t)/b$, so:

$$N^e = N^* \quad \text{when} \quad (v - a - t)/b = (v - a)/2b \Rightarrow t = (v - a)/2$$

Each driver imposes an external cost of b on $N^* = (v - a)/2b$ drivers.

By spending an amount X on road improvements, the government reduces the cost per trip to:

$$c(N, X) = a + bN - gX$$

Assume there is no congestion charge.

- (e) Write down the new expression for the social surplus, W (including the cost, X , incurred by the government), when there are N drivers.

Find the new equilibrium traffic flow, N_e , and the equilibrium social surplus. Comment on the social value of road improvements. [20%]

$$W = N(v - a - bN + gX) - X.$$

$$v - a - bN + gX = 0 \Rightarrow N_e = (v - a + gX)/b. \quad \text{In equilibrium } W = -X.$$

Road improvements cannot raise welfare when drivers can enter freely. The better road increases traffic until total benefits = total private costs; social welfare is decreased by the spending on improvements. (Of course, road improvements could be beneficial if combined with an optimum toll.)

Part B

11. An economy ('Home') has a fixed endowment of labour, L , which it can employ in two sectors with outputs x and y . The sectors have production functions:

$$x = L_x^{1/2}, \quad y = L_y^{1/2}$$

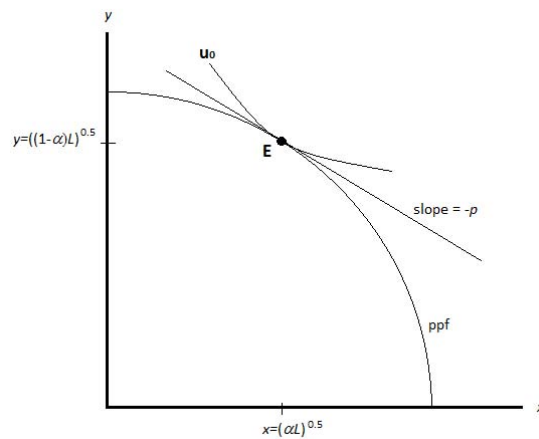
where L_x and L_y are the labour inputs in the two sectors, and $L_x + L_y = L$.

- (a) Derive, and sketch, the production possibility frontier of the economy. What is the marginal rate of transformation between x and y ? [15%]

The resource constraint is $L_x + L_y = L$. Rewrite in terms of x and y :

$L_x = x^2, L_y = y^2 \Rightarrow$ the ppf is $x^2 + y^2 = L$.

The MRT is the slope of the ppf: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \text{MRT} = \frac{dy}{dx} = -\frac{x}{y}$.



The economy contains a single consumer with utility function $U(x, y) = x^\alpha y^{1-\alpha}$, where $0 < \alpha < 1/2$. The consumer receives the total income of the economy, in the form of wages and profits.

- (b) What is the marginal rate of substitution between x and y ? [5%]

MRS = slope of indifference curve, dy/dx :

$$\text{MRS} = -\frac{\partial U}{\partial x} / \frac{\partial U}{\partial y} = -\frac{\alpha}{1-\alpha} \frac{y}{x}$$

- (c) Let the price of good y be 1. Find the competitive equilibrium price, p , of good x , and the equilibrium quantities of the two goods. Explain how and why the quantities produced depend on the parameter α . Illustrate the equilibrium on your diagram. [30%]

At the equilibrium, $p = \text{MRS} = \text{MRT}$:

$$p = \frac{\alpha}{1-\alpha} \frac{y}{x} = \frac{x}{y} \Rightarrow p = \frac{x}{y} = \sqrt{\frac{\alpha}{1-\alpha}}$$

The equilibrium quantities satisfy $x^2 + y^2 = L$ and $x = py$:

$$(p^2 + 1)y^2 = L \Rightarrow y = \sqrt{\frac{L}{1+p^2}}; \quad x = \sqrt{\frac{p^2 L}{1+p^2}} \Rightarrow y = \sqrt{(1-\alpha)L}; \quad x = \sqrt{\alpha L}$$

$\alpha < 0.5$, so consumers demand more y than x , and the price of x , $p < 1$. So more labour is employed in the y -sector (a fraction $(1-\alpha)$ of the labour endowment L), and more y is produced.

In the diagram, equilibrium E is on the ppf where the slope is equal to the slope of the indifference curve, and the price p . The economy produces more y than x .

Another economy ('Foreign') has the same technology, and but it has labour endowment L^* , and its consumer has a (different) utility function $U(x, y) = x^{1-\alpha}y^\alpha$. The two countries trade freely.

Suppose that $L = L^*$.

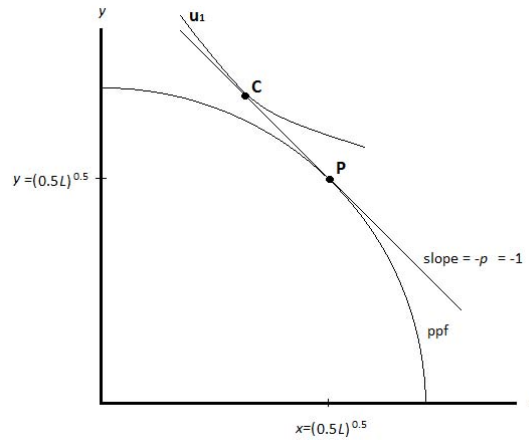
- (d) Show that the world price ratio, p , must be 1, explaining your reasoning carefully. [15%]

Both countries have the same resources, and in the free-trade equilibrium, they both face the same price, so they will have the same production ($x = x^*$ and $y = y^*$), and hence the same income, $m = px + y$. With Cobb-Douglas preferences the home expenditure on x is $(1-\alpha)m$ and foreign expenditure on x is αm . Total expenditure on x is m ; and similarly total expenditure on y is m . x and y are produced with the same production functions, so by symmetry $x = y$, and hence $p = 1$.

- (e) What is the pattern of trade between the countries? [10%]

Home has a higher demand for y , and Foreign for x ; production is the same in both. So Home exports x and imports y , and Foreign vice-versa.

- (f) Are there gains from trade? Explain and illustrate your reasoning carefully.
(A formal proof is not required.) [10%]



In the diagram production is at P and the budget constraint of the Home consumer is a line through P of slope $-p$. Consumption is at C, where $MRS = -p$. Since the price is different from the price under autarky, there are gains: the consumer can move beyond the ppf, and C is on a higher indifference curve (u_1) than point E under autarky (u_0).

- (g) What happens to the world price ratio, p , if L^* increases? What happens to real income in the Home country? [15%]

If foreign resources increase, it will produce more, and have higher income, and higher demand for both goods, but the demand for x increases more than the demand for y . Hence p , the price of x , will rise; more x than y will be produced. Real income at Home (which exports x) rises (e.g. can see this from the diagram - original C is now strictly inside the budget set).